

Lwandiso Zwane

Machine Learning Engineer – Industrial AI & Reasoning Systems

South Africa • Open to Remote or Relocation

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[\[GitHub\]](#) • [\[Portfolio\]](#) • [\[LinkedIn\]](#)

Professional Summary

Highly motivated and **Certified Machine Learning Engineer** specialising in Industrial AI, Reasoning Systems, and MLOps. Proven ability to **design, architect, and deploy** complex, modular microservice systems (FastAPI, Docker, AWS) for predictive maintenance and real-time anomaly detection. Focused on building **auditable and scalable** AI solutions, demonstrated through rigorous, self-directed projects (GridGuard, AVA) utilising multi-agent orchestration and modern RAG frameworks.

Technical Skills

Machine Learning & Reasoning Systems: Predictive Maintenance, Anomaly Detection, NLP Intent Parsing, Sequence Models (LSTM/BI-LSTM), Model Evaluation, Feature Engineering, AI Agents, LLMs.

Backend & Systems: Python (Expert Proficiency), FastAPI, Node.js, REST APIs, Microservice Architecture, Modular Reasoning Pipelines, MySQL, MongoDB, Pinecone.

Cloud & DevOps: AWS (Lambda, S3, EC2), GCP (BigQuery, Cloud Run), Docker, CI/CD (GitHub Actions), Terraform

Data & Tooling: Pandas, NumPy, Scikit-learn, SQL (PostgreSQL, MySQL), Logging Systems, FTIR/FDIR Tracing

Projects

[AVA – Modular Predictive-Maintenance Reasoning Engine \(2025\)](#)

- **Tech Stack:** Python, Microservices, Modular RAG, LLM Plugin Agents

- Built a custom Retrieval-Augmented Generation (RAG) framework to turn raw industrial telemetry into actionable insights.
- Integrated LLM-based plugin agents for anomaly analysis, adaptive tool recommendation, and dynamic reasoning.
- Developed machine-agnostic profiles enabling multi-agent orchestration and context-aware decision-making, **reducing diagnostic latency by 35% in testing environments.**
- Implemented full FTIR-style trace logging, making every reasoning step auditable and transparent.

GridGuard – Smart-Grid Diagnostic Reasoning Engine (2025)

- **Tech Stack:** Python, Microservices, RAG, Multi-Agent Coordination
- Extended AVA to energy grids: multi-agent orchestration over live and simulated telemetry, combining automated reasoning with adaptive diagnostics.
- Created contextual memory tied to machine profiles to track historical faults and predict potential failures, **resulting in a simulated 15% reduction in unplanned downtime.**
- Enabled dynamic tool execution where LLM agents select, coordinate, and adapt diagnostic routines based on prior outcomes.
- Demonstrated end-to-end flow: sensor input → intent → reasoning → tool execution → actionable report.

Industrial NLP Intent Parser – Technician Command Understanding (2025)

- **Tech Stack:** DistilBERT, LSTM/Bi-LSTM, Hybrid Token Classification
- Designed an end-to-end intent extraction pipeline for industrial operating and diagnostic commands.
- Built a topic router + target/parameter extractor to normalize and map noisy inputs into structured actions.
- Implemented three model variants (BERT-based, BiLSTM, LSTM) and evaluated performance for accuracy and robustness, **achieving a final intent classification accuracy of 94.2%.**

Selected Achievements

- **Architected** full FTIR-style trace-logging across three independent industrial reasoning systems for auditable compliance.
- **Designed and implemented** three complex, independent industrial reasoning engines (GridGuard, AVA, NLP Parser) within a year.
- Developed fully modular architectures enabling microservice-level scalability and deployment per subsystem.

- **Validated** multiple production-ready demo environments showcasing end-to-end real-world diagnostic workflows.
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Education

Certified Machine Learning Engineer (MLE) Track – Codecademy (2023)

- **Certification Achieved** upon completion of intensive, project-based curriculum.
- Core competency in ML Fundamentals, Software Engineering principles, Algorithms, Pandas, and Scikit-learn.

Supplementary Technical Courses

React Native (2021), Node.js (2021), Cloud Computing for ML – IBM/Coursera (Audit)